training the immune system to fight the virus and preparing the body sufficiently for future protection. In order to achieve this, a strain of the virus is inactivated using chemicals and converted to a safe immunogen and administered with an adjuvant (a substance used to improve the immune response) in multiple doses. This is one of the more traditional platforms on which several vaccines for other pathogens have been developed successfully in the past.

SII's Covishield is based on a viral vector platform. The SARS-CoV-2 virus is studded with proteins which it then uses to enter human cells. Covishield works by targeting these proteins by using double stranded DNA. The vaccine adds the gene for the coronavirus spike protein to a modified virus derived from a chimpanzee adenovirus which can enter human cells but can't replicate inside them. On being injected, the adenoviruses latch on to the proteins on the cells' surface from which they are engulfed. Once engulfed, they release the DNA into the nucleus of the cells from which the gene for the coronavirus spike protein can be read by the cell and copied into a molecule called mRNA which further leads to formation of spikes which can be recognized by the immune system and generate an immune response. This platform has been under research for years and was only recently approved for an Ebola vaccine made by Johnson & Johnson. Using a DNA strand makes the vaccine more tolerant and the adenovirus' tough protein coat helps protect the genetic material inside. This means it is less fragile than those based on RNA platforms, doesn't need to be frozen and has a longer shelf life when refrigerated at 2–8 degree Celsius (normal refrigerator temperature), making it logistically easier to transport and store in a vast country like India.

The Zydus Cadila vaccine, on the other hand, uses a new technology platform called DNA plasmid. It uses small rings of DNA or plasmids containing genetic information which is carried to the cells in order to make 'spike proteins', which can be used to penetrate the cells and generate an immune response. Since it is based on DNA, it is also stable at higher temperatures and doesn't need to be frozen, making logistics easier. This is also a needle-free vaccine with less adverse side effects, making it safer for administration in children. The vaccine can also be manufactured in laboratories which are Biosafety level I versus those based on inactivated virus platform which need Biosafety level III manufacturing units, thus making it relatively easier and cheaper to achieve large-scale production